

# Active Self-Correcting 3D Intelligence

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Research Proposal · PhD Application

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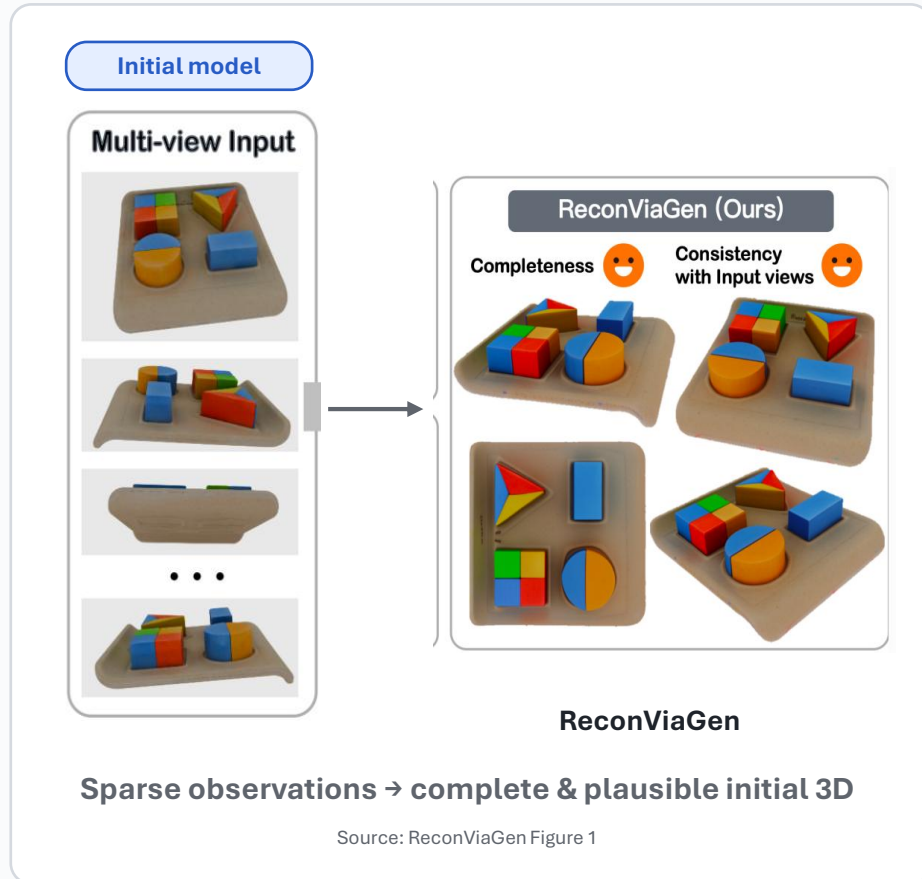
**Spatial intelligence that actively uses observations to verify and refine an explicit 3D object model so that it progressively matches the real object.**



**Near-term**  
**Rigid object geometry self-correction**

**Long-term**  
**Accurate 3D modeling of changing object states**

# From plausible reconstruction to self-correcting precision



## Missing next step

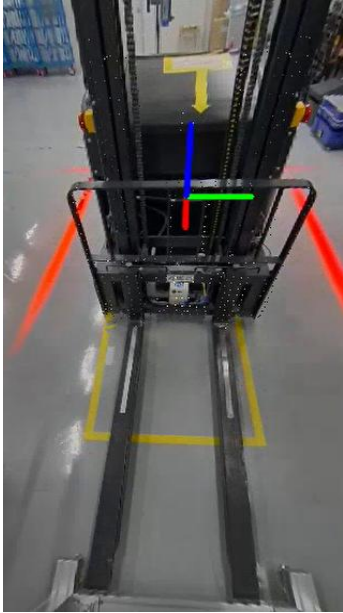
**High-precision instance modeling needs more than plausible completion.**

**We need an agent that can:**

- 1. check whether its model matches real observations**
- 2. gather more evidence to identify the cause of a mismatch**
- 3. update the part of the model that is wrong**

# Core gap: What is causing the mismatch?

Local tracking



Real observation + 6D pose | internal mesh

Source: my FoundationPose local result

**Hypothesis A**  
pose is wrong

**Hypothesis B**  
geometry is wrong

Pose trackers primarily solve alignment.  
A remaining mismatch may come from pose error or  
model geometry error.

→ First diagnose the cause, then correct the model

AREA3D<sup>1</sup>: uncertainty · ActivePose<sup>2</sup>: pose ambiguity · FvOR<sup>3</sup>: joint refinement

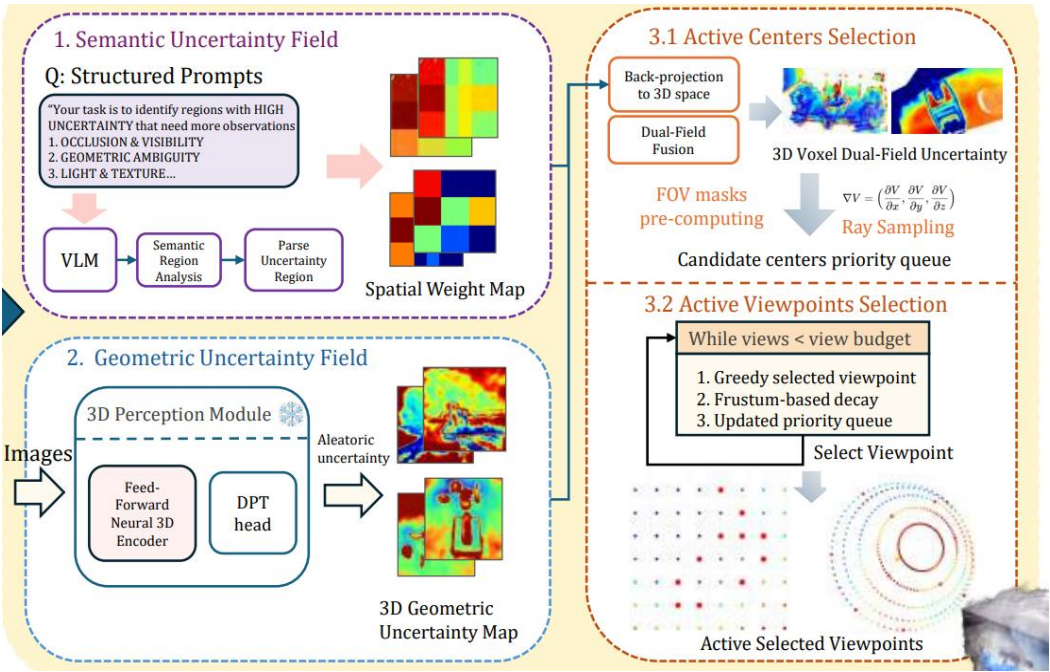
**Ours: error diagnosis**

1. [AREA3D: Active Reconstruction Agent with Unified Feed-Forward 3D Perception and Vision-Language Guidance](#)
2. [ActivePose: Active 6D Object Pose Estimation and Tracking for Robotic Manipulation](#)
3. [FvOR: Robust Joint Shape and Pose Optimization for Few-View Object Reconstruction](#)

# First Research Question: Where should the agent look next?

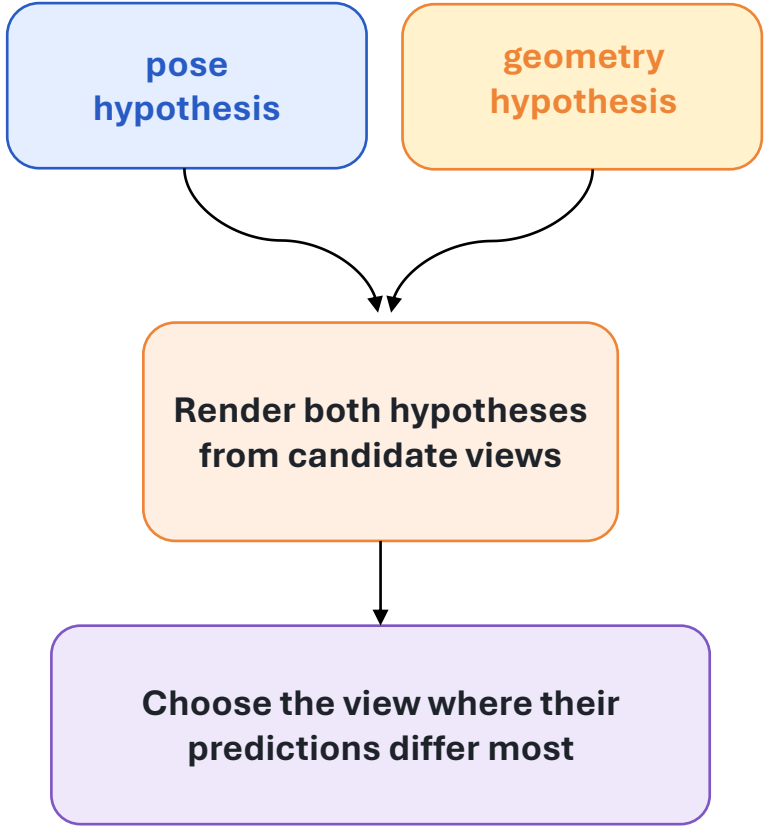
Which next view best distinguishes pose error from local geometry error?

## Existing active reconstruction



AREA3D

Observe uncertain geometry  
Reduce reconstruction uncertainty



Ours

Observe where error explanations diverge  
Diagnose the source of mismatch

# Technical building blocks for the diagnostic loop

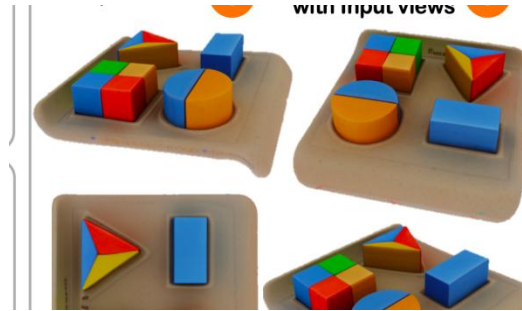
Missing link:

Diagnose the error → Choose the right view

→ Correct the 3D model

**Proposed contribution: a closed loop that connects these capabilities.**

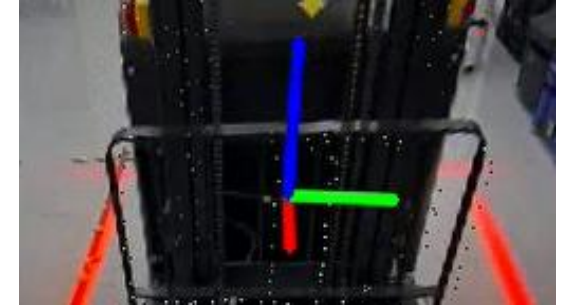
## Initial 3D



**Build an initial 3D model**

ReconViaGen

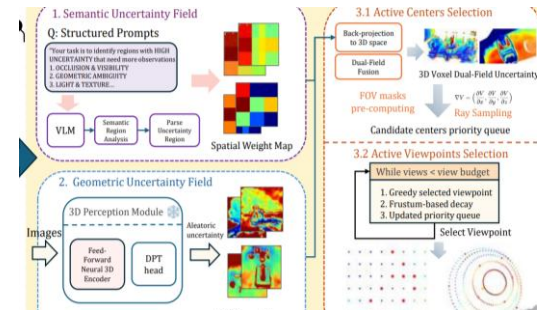
## Tracking



**Track the object pose**

FoundationPose local result

## Active view



**Choose informative views**

AREA3D

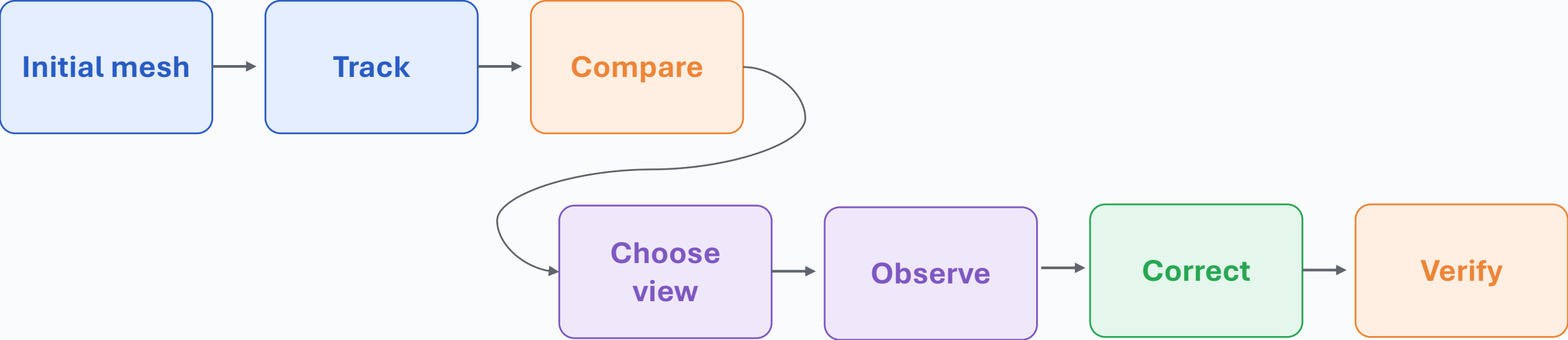
## Editable 3D



**Edit the 3D representation**

Radiance Meshes

# Proposed first step: a minimal closed loop



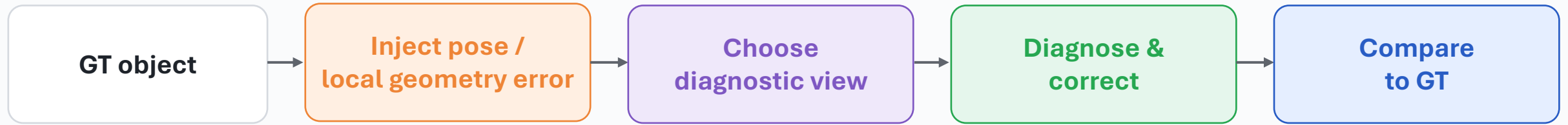
## Core idea

Hypothesis-driven diagnostic view selection  
Compare pose-error and geometry-error explanations,  
then observe where their predictions differ most.

## Scope control

Primary target = geometry  
Appearance/texture = evidence + output consistency

# Evaluation plan: controlled benchmark



## Metrics

- diagnosis accuracy: pose vs local geometry
- local geometry error reduction
- views needed to reach target accuracy
- tracking improvement after correction

## Baselines

- fixed views / random orbit
- generic uncertainty-based NBV
- no diagnostic selection
- no geometry update

**Start: Isaac Sim + rigid objects + synthetic defects → then real RGB-D / RGB-only extension**

# Long-term roadmap

## Representation fidelity

**Geometric fidelity**

static geometry self-correction

**Appearance fidelity**

geometry + texture consistency

**State fidelity**

track the real object's changing 3D state

## Active evidence

**Visual observation**

**Active viewpoint selection**

**Physical interaction**

**Manipulation as sensing**

**Goal: Keep the explicit 3D object model progressively aligned with the real object's current state.**